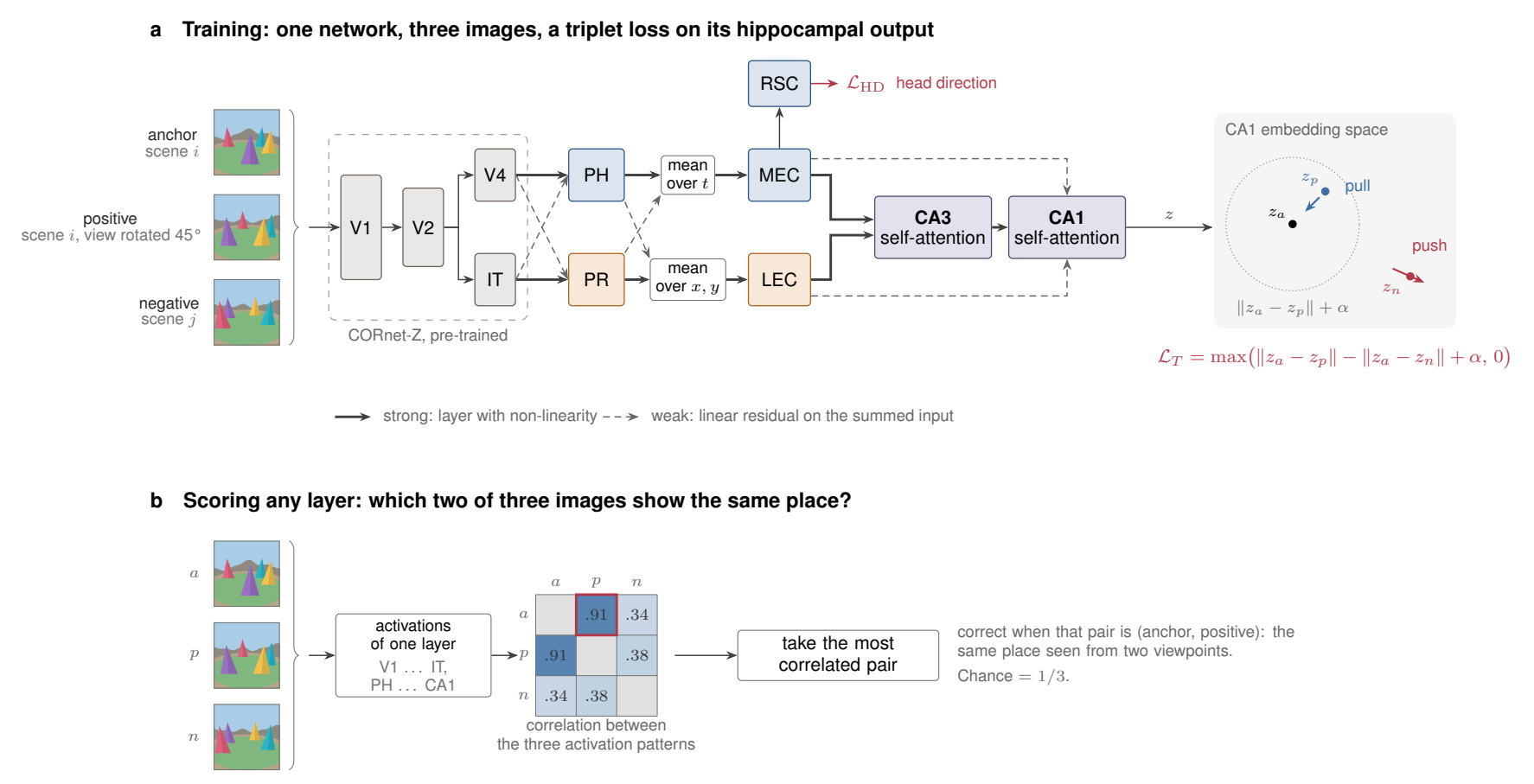

a Training: one network, three images, a triplet loss on its hippocampal output
anchor
scene i
positive
scene i, view rotated 45°
negative
scene j
V1
V2
V4
IT
PH
PR
mean over t
mean over x, y
RSC
$\mathcal{L}_{\mathrm{HD}}$ head direction
MEC
LEC
CA3
self-attention
CA1
self-attention
z
CORnet-Z, pre-trained
CA1 embedding space
z_p
pull
z_a
push
z_n
$\|z_a - z_p\| + \alpha$
$\mathcal{L}_T = \max(\|z_a - z_p\| - \|z_a - z_n\| + \alpha, 0)$
strong: layer with non-linearity
weak: linear residual on the summed input
b Scoring any layer: which two of three images show the same place?
a
p
n
activations
of one layer
V1 … IT,
PH … CA1
.91
.34
.91
.38
.34
.38
correlation between
the three activation patterns
take the most
correlated pair
correct when that pair is (anchor, positive): the same place seen from two viewpoints.
Chance $= 1/3$.